



UNIVERSITY OF GHANA

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DEPARTMENT OF COMPUTER SCIENCE
SCHOOL OF PHYSICAL AND MATHEMATICAL SCIENCES

SECOND SEMESTER 2025/2026 ACADEMIC YEAR

COURSE SYLLABUS

Course Code and Title: DCIT 418 System and Network Security

Credits: Three (3) Credits

Lecture Period(s) and Venue: See timetable

Prerequisites: NIL

Course Instructor

- Name: Dr Edward Danso Ansong
- Office Location: Department of Computer Science, R10
- Office Hours: Thursday 2-4pm;
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Primary Book: *Cryptography and Network Security – Principles and Practice* (William Stallings, 6th Ed.)

The syllabus below is designed for undergraduate Computer Science / Information Technology programme with a strong practical component. It combines theory, laboratory exercises, cryptographic programming, packet analysis, and security demonstrations.

DCIT 418: Systems and Network Security

Semester Course Syllabus (13 Weeks)

Course Description

This course introduces students to the principles and practices of systems and network security. It covers computer security concepts, security attacks and services, classical and modern cryptographic techniques, symmetric encryption systems, block cipher operations, pseudorandom number generation, stream ciphers, and the mathematical foundations of cryptography. Students gain practical experience implementing cryptographic algorithms, analyzing security attacks, and evaluating secure communication systems. The course is based primarily on Chapters 1–8 of William Stallings' *Cryptography and Network Security: Principles and Practice* (6th Edition).

Course Learning Outcomes

By the end of the semester, students should be able to:

1. Explain fundamental concepts of computer and network security.
2. Analyze security threats, attacks, services, and mechanisms.
3. Implement and evaluate classical cryptographic techniques.
4. Explain the operation and security properties of DES and AES.
5. Apply modular arithmetic and finite field mathematics in cryptography.
6. Analyze block cipher modes of operation and their security implications.
7. Evaluate pseudorandom number generators and stream ciphers.
8. Apply number theory concepts used in modern cryptography.
9. Develop simple cryptographic applications using Python.
10. Conduct basic security assessments and cryptographic experiments.

Semester Schedule (13 Weeks)

Week	Topic / Theme	Lecture Focus	Lab Activities & Tools	Deliverables	Reading
1	Introduction to Systems and Network Security	Course overview, CIA Triad, Security Goals, Security Challenges, Security Architecture	Introduction to Kali Linux, Ubuntu Security Tools, Wireshark Installation	Lab Setup Report	Chapter 1
2	Security Attacks, Services and Mechanisms	Passive vs Active Attacks, Security Services, Security Mechanisms, OSI Security Architecture	Wireshark packet capture and attack identification	Security Attack Analysis Report	Chapter 1

3	Classical Cryptography I	Symmetric Cipher Model, Caesar Cipher, Monoalphabetic Substitution, Cryptanalysis	Python implementation of Caesar and Substitution Ciphers	Assignment 1	Chapter 2
4	Classical Cryptography II	Vigenère Cipher, Playfair Cipher, Transposition Techniques, Rotor Machines, Steganography	Python Cryptography Toolkit; OpenStego	Lab Report 1	Chapter 2
5	Block Cipher Fundamentals and DES	Block Cipher Principles, Feistel Structure, DES Architecture, DES Security	DES Encryption/Decryption using Python and OpenSSL	DES Implementation Exercise	Chapter 3
6	Number Theory Foundations	Divisibility, Euclidean Algorithm, Modular Arithmetic, Groups, Rings and Fields	SageMath/Python Modular Arithmetic Exercises	Assignment 2	Chapter 4
7	Finite Fields and AES Mathematics	Polynomial Arithmetic, $GF(p)$, $GF(2^n)$, Finite Field Operations	SageMath Finite Field Computations	Lab Report 2	Chapter 4
8	Advanced Encryption Standard (AES)	AES Structure, SubBytes, ShiftRows, MixColumns, Key Expansion	AES Implementation using Python Cryptography Libraries	AES Practical Exercise	Chapter 5
9	AES Applications and Mid-	AES Security Analysis, AES Performance,	AES Performance and Testing Benchmarking	Mid-Semester Test / Quiz	Chapter 5

	Semester Assessment	Implementation Considerations			
10	Block Cipher Modes of Operation	ECB, CBC, CFB, OFB, CTR, Triple DES, Storage Encryption	OpenSSL and Python Demonstrations of Modes	Assignment 3	Chapter 6
11	Pseudorandom Number Generation	PRNG Principles, Randomness Testing, Block-Cipher-based PRNGs	Random Number Generation and Entropy Analysis using Python	Lab Report 3	Chapter 7
12	Stream Ciphers and RC4	Stream Cipher Design, RC4 Operation, Security Issues in Stream Ciphers	RC4 Implementation and Security Evaluation	Mini Project Submission	Chapter 7
13	Number Theory for Public-Key Cryptography	Prime Numbers, Euler's Theorem, Fermat's Theorem, Chinese Remainder Theorem, Discrete Logarithms	SageMath Cryptographic Mathematics Workshop	Final Project Presentation	Chapter 8

Teaching Methodology

This course adopts a **practice-oriented and hands-on teaching approach**, emphasizing the application of systems and network security concepts through laboratory exercises, simulations, cryptographic implementations, and security analysis activities. While theoretical concepts provide the foundation, the primary focus of the course is on enabling students to develop practical cybersecurity and cryptographic skills relevant to modern computing environments.

Teaching will be delivered through a combination of:

1. Interactive Lectures

- Introduction of core concepts, principles, and security models.
- Guided discussions on security threats, vulnerabilities, and mitigation techniques.
- Case studies of real-world cybersecurity incidents and attacks.

2. Hands-on Laboratory Sessions

- Weekly practical exercises involving cryptographic algorithms, encryption systems, and network security tools.
- Packet analysis, network traffic monitoring, and security assessments.
- Implementation of cryptographic techniques using Python and open-source security tools.

3. Demonstrations and Simulations

- Instructor-led demonstrations of security attacks and defensive mechanisms.
- Simulated security environments for understanding attack vectors and countermeasures.

4. Project-Based Learning

- Students will complete individual or group projects that require the design, implementation, or evaluation of security solutions.
- Projects will encourage problem-solving, critical thinking, and practical application of concepts learned throughout the semester.

5. Collaborative Learning

- Group discussions and peer-learning activities.
- Analysis of emerging cybersecurity trends, standards, and technologies.

6. Independent Study and Research

- Reading assignments from the prescribed textbook and supplementary materials.
- Exploration of current cybersecurity tools, vulnerabilities, and best practices.

Practical Component: Approximately **60–70% of course contact hours** will be devoted to laboratory work, demonstrations, programming exercises, and security tool usage, while **30–40%** will focus on theoretical foundations and conceptual understanding.

Attendance Policy

Regular attendance and active participation are essential for success in this course due to its strong practical and laboratory-based nature.

1. Students are expected to attend all lectures, tutorials, and laboratory sessions.
2. Attendance will be taken during both lectures and practical sessions randomly.
3. Students who miss laboratory sessions are responsible for obtaining missed materials and completing any assigned exercises within the stipulated time.
4. Absences due to medical emergencies, official university assignments, or other approved reasons must be supported by appropriate documentation and communicated promptly to the lecturer.
5. Repeated lateness or unexplained absences may result in reduced participation marks and may affect eligibility for practical assessments.
6. Because laboratory exercises build progressively throughout the semester, students are strongly encouraged to maintain consistent attendance.

Academic Integrity Policy

Students are expected to uphold the highest standards of academic honesty, professionalism, and ethical conduct throughout the course.

1. All submitted assignments, laboratory reports, projects, and examinations must represent the student's own work unless collaboration is explicitly permitted.
2. Plagiarism, including copying material from books, websites, artificial intelligence tools, or other students without proper acknowledgment, is strictly prohibited.
3. Unauthorized sharing, copying, or distribution of laboratory reports, assignments, source code, or project solutions constitutes academic misconduct.
4. Any form of cheating during quizzes, tests, practical assessments, or examinations will be handled according to university regulations and disciplinary procedures.
5. Students may discuss concepts and problem-solving approaches with classmates; however, submitted work must be independently produced unless otherwise specified.
6. Proper citation and referencing must be used when incorporating information, code snippets, datasets, research papers, or external resources.
7. The use of cybersecurity tools and techniques taught in this course must be limited to authorized educational and research environments. Students are responsible for any action of unauthorized access to systems, networks, or data that are not strictly prohibited and may result in disciplinary action.

8. Students are expected to demonstrate ethical behaviour consistent with professional cybersecurity standards and practices.

Recommended Laboratory Environment

Category	Tools
Operating Systems	Ubuntu Linux, Kali Linux
Programming	Python 3.x
Cryptography Libraries	PyCryptodome, Cryptography
Mathematics	SageMath
Network Analysis	Wireshark
Security Testing	Nmap
Encryption Utilities	OpenSSL
Steganography	OpenStego
Development Environment	VS Code

Assessments

Assessment	Weight
Assignments (3)	10%
Laboratory Reports (3)	10%
Mid-Semester Test	10%
Mini Project	15%
Quizzes	5%
Final Examination	50%
Total	100%

Semester Mini Project Options

Students may choose one of the following:

1. Design and implement a secure file encryption system using AES.
2. Develop a classical cipher toolkit supporting multiple algorithms.
3. Create a cryptographic key generator and randomness analyzer.

4. Implement a secure messaging prototype using symmetric encryption.
5. Analyze and compare block cipher modes using experimental data.
6. A lot more will be added.

Academic Integrity Statement

"Students enrolled in DCIT 418: Systems and Network Security are expected to conduct themselves in a manner consistent with the ethical principles of the cybersecurity profession. Any form of academic dishonesty, plagiarism, unauthorized system access, or misuse of security tools will be treated as a serious violation of university policy and may result in penalties ranging from loss of marks to disciplinary action."